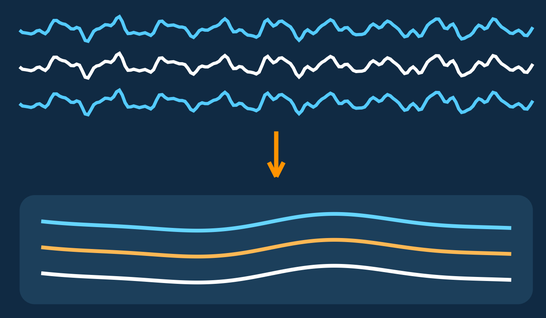
EEG TO BOLD / COMPUTER SCIENCE PHD EXPLORATION

Can EEG recover clinically useful fMRI?

More data helped, but cross-site prediction remains weak. A useful PhD should target one clinical decision and measure when a model must abstain.



EXPLORATORY RESEARCH | MONASH M3 EXPERIMENTS | 21 OCTOBER 2026

THE CLINICAL QUESTION

One task. A measurable benefit.

EEG: scalp electrical activity.
BOLD: blood oxygenation in fMRI.
One useful clinical decision?

PUBLISHED MODEL REFERENCE

Inside UnEBOLT

Simplified architecture [1]; not reproduced here.

EEG history
Before each BOLD frame

↓

Shared EEG encoder
Temporal + spectral; gated fusion

↓

Regional adapters
ROI embeddings and output heads

↓

BOLD trajectories
One predicted series per region

Patient BOLD fidelity and clinical utility remain unvalidated.

01 / EVIDENCE FROM M3

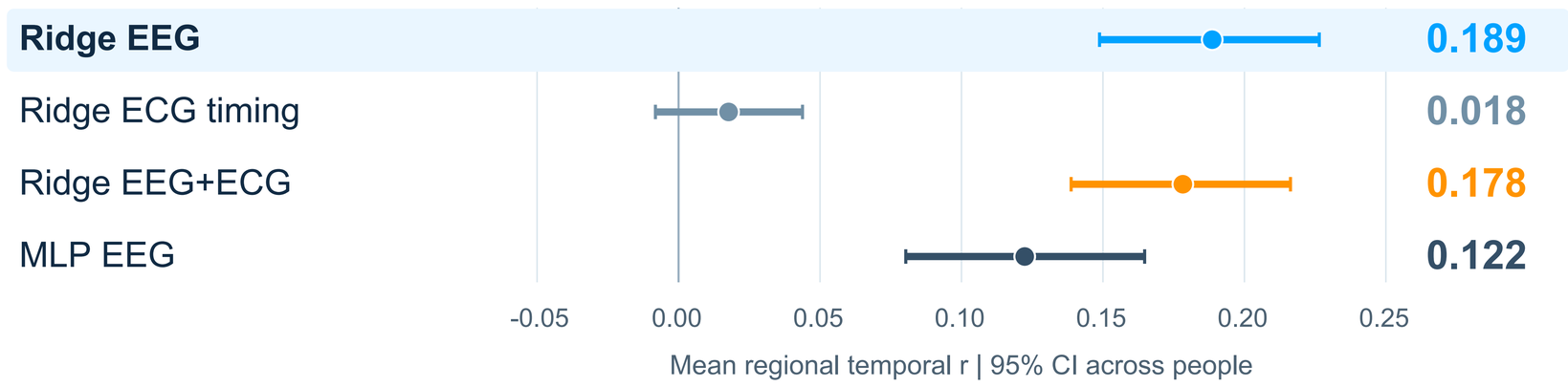
A linear baseline stays ahead

ds006040 development | EC-trained / EC-tested | 64 regional targets

27 held-out people

96 fitted experiment cells

300 matched EEG-power features



12 training people x 240 retained target-seconds/person. EEG ridge predictive R^2 = .028.

More data helped; extra complexity did not.

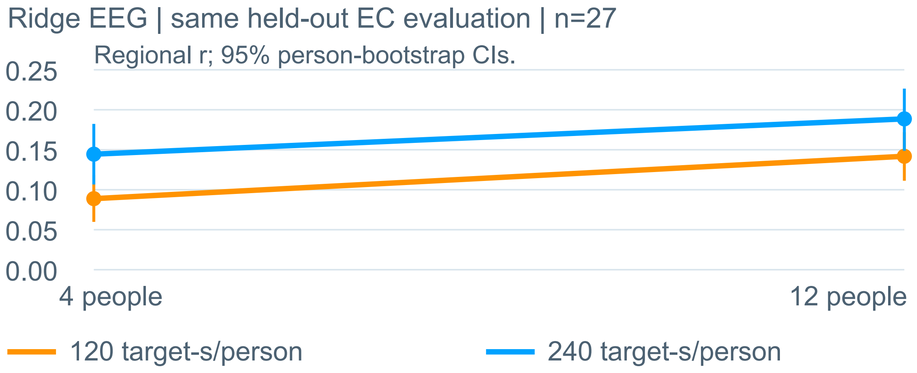
The small MLP and sparse ECG pipeline did not improve the EEG baseline. Regional gain beyond a predicted-global template remains uncertain: paired delta-r .004 (95% CI -.008 to .017). These tests do not establish a biological ceiling.

THE PHD OPPORTUNITY / reliable prediction across people and sites

02 / EFFECT OF TRAINING DATA

People and recording time help

Ridge EEG | same held-out EC evaluation | n=27



WHAT WAS ACTUALLY TESTED

ds006040: 27 people/54 runs; three subject-grouped outer folds, separate inner tuning. Ridge/small MLP; EEG, sparse ECG timing/availability, or both. Cross 4/12 training people and 120/240 retained target-seconds/person. Fixed subsets; one neural seed; four equal-weight MLP target groups. Published models were not reproduced.

EXTERNAL DATA / SECONDARY EXPLORATION

NATVIEW: transfer remains weak

Strict timing QC: 1/8. Amended QC: 6/8; n=6.
Metadata-only amendment; ingestion pilot excluded.

0.104 regional r

-0.071 predictive R^2

r: 95% CI [-0.0002, 0.207]; includes zero.

740/2304 frames scored (32.1% of all eight).
Frozen VU model/QC; 23 target sites include interpolation.

03 / THREE FALSIFIABLE THESIS QUESTIONS

Recover. Transfer. Know when to stop.

1. **Measure useful information**
   Which montage, EEG history and sensors add held-out value for their acquisition cost?
2. **Transfer structured dynamics**
   Can shared dynamics and counted calibration improve regional signal beyond ridge and state controls?
3. **Predict selectively**
   What error is achievable at useful coverage under specified shift? Compare abstention with quality rejection.

CUTTING-EDGE MODELS / PROPOSED USES

UnEBOLT [1] | 64 ROIs, r .418 | regional feedback target

UniEFS [2] | 512 ROIs, r .367 | Parkinson classification probe

MARBLE [3] | 64 ROIs, r .434; 2.6M parameters | compact estimation

Healthy temporal r, different protocols: not a ranking or patient validation.

APPLICATIONS / VALIDATION GATES

First route: wakefulness advisory; compare direct EEG arousal prediction.
Second route: one regional feedback target; require paired patient validation and controlled benefit.
Precedent: Prism is cleared for adjunctive PTSD biofeedback; diagnostic interpretation excluded [4].
Limits: retrospective preprocessing; normalized targets, no absolute %BOLD calibration; sparse ECG, one development site/seed. Live use and clinical benefit untested [5].

[1] UnEBOLT, MIDL 2026. **[2]** UniEFS, ICML 2026.
[3] MARBLE, MICCAI 2026. **[4]** FDA K222101, 2023.
[5] DECIDE-AI, 2022. Full sources: accompanying report.

Seamus Taylor
PhD thesis exploration / Evidence and full methods in the accompanying report